

A Normalized Hénon Galois Norm at the Critical Boundary

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Abstract

We apply Galois descent to a conjugate-paired family of finite-field Hénon augmentation determinants whose trace fields have unbounded degree. The ordinary rational norm has virtual local degree $2(p-1)$ and therefore cannot come from a uniformly bounded-rank graded local operator. In contrast, the canonical field-degree-normalized logarithmic norm has first moment $-12/(p-1)$ and uniformly bounded higher moments. We prove that its Euler product converges locally uniformly and is holomorphic and nonzero in the half-plane $\operatorname{Re} s > 1/2$. This advances the rigorously controlled Hénon Euler domain to the open Riemann critical boundary without using zero data or fitted parameters. The normalized local roots remain analytic germs rather than ordinary rational or Fredholm determinants; continuation, a functional equation, and integral divisor multiplicities are left as explicit gates.

1 Introduction

The coefficient-field obstruction for the full μ_3 -graded Hénon kernel is severe: at every split prime p , the conjugate-paired first moment generates $\mathbf{Q}(\zeta_p)^+$, of degree $(p-1)/2$. A conventional compatible system over one fixed number field is therefore impossible. Rather than choosing a different field independently at every prime, we test Galois trace and norm, the two minimal invariant descents.

The additive trace collapses the first moment to -6 . The multiplicative norm preserves the complete local chronological determinant and restores rational coefficients, but its virtual degree grows linearly with p . The central observation of this paper is that dividing the canonical logarithm by the trace-field degree preserves the ordered moments while adding exactly enough first-order decay to move the Euler convergence boundary.

Theorem 1.1 (Critical-boundary germ). *Let $G_p(z)$ be the origin-normalized field-degree root of the Galois norm of the conjugate-paired Hénon augmentation factor, as defined in Section 3. Then*

$$\mathcal{G}(s) = \prod_{p \equiv 1 \pmod{3}} G_p(p^{-s})$$

converges locally uniformly and is holomorphic and nonzero on $\operatorname{Re} s > 1/2$.

The statement is both stronger and more limited than an ordinary Euler product theorem. It reaches the critical boundary from the right, but the local factors are defined by a normalized logarithm on the unit disk. Their divisor orders need not be integral after division by $(p-1)/2$. Thus this paper proves an analytic determinant germ, not a finite-dimensional rational determinant or a Fredholm determinant.

Normalized determinants are natural in finite and von Neumann algebras [2, 1]; our construction does not yet claim such an operator realization. That distinction becomes the next theorem gate rather than a semantic choice.

2 Rational norm descent

Fix a prime $p \equiv 1 \pmod{3}$ and set

$$K_p = \mathbf{Q}(\zeta_p), \quad L_p = K_p^+, \quad d_p = [L_p : \mathbf{Q}] = (p-1)/2.$$

For the additive character $\psi_a(x) = \zeta_p^{ax}$, denote the C43 local augmentation factor by $D_{p,a}^{\text{aug}}(z)$. It is a ratio of sector characteristic polynomials of unitary matrices, has value one at zero, and has finite zero and pole support on $|z| = 1$.

Pair the inverse characters:

$$E_p(z) = D_{p,1}^{\text{aug}}(z) D_{p,-1}^{\text{aug}}(z) \in L_p(z).$$

Complex conjugation exchanges the factors. Every $\sigma \in \text{Gal}(L_p/\mathbf{Q})$ replaces the additive character while leaving the chronological phase and matrix-power convention unchanged. Consequently

$$\text{Log}_0 E_p(z) = - \sum_{n \geq 1} \frac{B_{p,n}}{n} z^n, \quad B_{p,n} \in L_p.$$

Define the ordinary rational norm

$$N_p(z) = \text{Norm}_{L_p/\mathbf{Q}} E_p(z).$$

Then $N_p(0) = 1$ and

$$\text{Log}_0 N_p(z) = - \sum_{n \geq 1} \frac{C_{p,n}}{n} z^n, \quad C_{p,n} = \text{Tr}_{L_p/\mathbf{Q}} B_{p,n}.$$

Proposition 2.1 (Growing virtual rank). *The ordinary norm has*

$$\text{vdeg } N_p = 4d_p = 2(p-1).$$

If $Q_p \in \mathbf{Q}(z)$ satisfies $|\text{vdeg } Q_p| \leq M$ independently of p , then

$$|\text{vdeg}(Q_p N_p)| \geq 2(p-1) - M.$$

Hence no uniformly bounded-rank family of finite-dimensional graded determinants can realize the ordinary norm.

Proof. Each paired conjugate has virtual degree four. Virtual degree is the order at infinity, hence is additive under products and unaffected by cancellation in a displayed numerator and denominator. The triangle inequality gives the second statement. $\square$

The first rational trace is exact:

$$C_{p,1} = -6. \tag{1}$$

It follows from the C44 zero-fiber identity $\#\{2x^3 + 2y^3 + (1+\rho)xy = 0\} = p-3$. Equation (1) does not imply that $(1-z)^{-6}$ is a complete local factor; higher moments are different.

3 The normalized logarithmic root

On $|z| < 1$, all local sector determinants are nonzero. There is therefore a unique analytic logarithm $\text{Log}_0 N_p$ with value zero at the origin. Put

$$G_p(z) = \exp(d_p^{-1} \text{Log}_0 N_p(z)), \quad G_p(0) = 1. \tag{2}$$

Its logarithmic moments are

$$\log G_p(z) = - \sum_{n \geq 1} \frac{c_{p,n}}{n} z^n, \quad c_{p,n} = C_{p,n}/d_p.$$

In particular,

$$c_{p,1} = -\frac{6}{d_p} = -\frac{12}{p-1}. \quad (3)$$

The smooth-leading cubic Artin–Schreier estimate inherited from C43 gives, for every embedding of the paired moment,

$$|B_{p,n}^\sigma| \leq 4 \cdot 4^n.$$

Averaging over all d_p embeddings yields

$$|c_{p,n}| \leq 4 \cdot 4^n. \quad (4)$$

Equations (3) and (4) have different analytic roles. The exceptional first moment has an extra factor p^{-1} ; the uniform bound starts contributing only at $n = 2$. Their intersection is the half-plane in Theorem 1.1. The detailed normal-convergence proof is in Appendix A.

Remark 3.1. No parameter is adjusted to place the boundary at $1/2$. The exponent is forced by the exact trace-field degree and the fact that the first nontrivial uncontrolled repetition is $n = 2$.

4 Chronological second-moment control

The four-variable phase for the second determinant moment is

$$\Phi_{p,2} = 2 \sum_{j=0}^3 x_j^3 + x_0 x_1 + x_1 x_2 + x_2 x_3 + \rho x_3 x_0.$$

This is the exact ordered two-step-kernel repetition. It is not a product of averaged transition matrices. Let $Z_{p,2} = \#\Phi_{p,2}^{-1}(0)$. Cyclotomic trace gives

$$C_{p,2} = \frac{2Z_{p,2}}{p} - 2p^2, \quad c_{p,2} = \frac{2C_{p,2}}{p-1}.$$

p	7	13	19	31	37	43	61	67	73	79	97
$C_{p,2}$	-6	-6	-6	-30	18	-54	18	42	-30	42	-30

The ledger proves that canceling the first coefficient with $(1-z)^6$ is not an all-order Tate cancellation. It does not prove a distribution law or a natural boundary.

Over characteristic zero with $\rho^2 + \rho + 1 = 0$, the affine singular ideal of $\Phi_{p,2}$ has only the origin, and its projective leading cubic is smooth. Thus the second moment is a legitimate fixed-dimensional cohomological problem; irregular finite-prime values should not be promoted to random-noise evidence.

5 Determinant and boundary firewall

The function G_p in (2) is canonically single-valued on the open unit disk. Suppose N_p has divisor valuation m at a boundary point. Analytic continuation of the chosen root has local order m/d_p . A single-valued meromorphic function requires this order to be integral. Consequently an ordinary rational root can exist only if every divisor valuation of N_p is divisible by d_p .

This exact divisibility condition separates three statements:

1. the normalized logarithmic germ exists on $|z| < 1$;
2. an ordinary rational or Fredholm determinant would require additional integral-divisor and operator theorems;
3. global continuation across $\operatorname{Re} s = 1/2$ would additionally require control of cross-prime divisor coincidences and monodromy.

Only the first is proved here. Under the optional display $z = p^{1/2-s}$, local unit-circle divisors lie on $\operatorname{Re} s = 1/2$, but this geometric observation is not a natural-boundary theorem. HCS-C46 tests the divisor-integrality condition exactly at the first split prime.

6 Route-A evaluation and conclusion

The normalized norm has exact Hénnon chronological moments and a locally uniform analytic Euler determinant germ. Its domain reaches the open critical boundary. The primes remain arithmetic fibers rather than primitive orbits of one real system, and no continuation, Gamma factor, functional equation, or Hilbert–Pólya operator is known.

The strict evaluation is

$$\begin{aligned} A1 &= \text{WEAK}, & A2 &= \text{ANALYTIC_DETERMINANT}, \\ A3 &= \text{PARTIAL_ANALYTIC_STRUCTURE}, & A4 &= \text{NATURAL_QUANTIZATION}. \end{aligned}$$

Overall the project is an exploratory critical-boundary germ. It does not authorize Route B.

The result changes the direction of the program. Fixed-field descent is impossible and ordinary norm has growing virtual rank, yet the canonical normalized trace does not collapse: it improves the analytic domain. The next two gates are therefore structural rather than numerical. First decide whether local normalized roots are ordinary determinants; if not, construct the normalized-trace operator category in which their fractional dimensions are legitimate.

A Normal convergence

Let K be compact in $\operatorname{Re} s > 1/2$, and choose $\sigma_0 > 1/2$ with $\operatorname{Re} s \geq \sigma_0$ on K . Every fixed local factor is analytic because $|p^{-s}| < 1$. We may remove finitely many primes so that $4p^{-\sigma_0} \leq 1/2$ in the remaining tail.

The $n = 1$ contribution to the logarithm is bounded uniformly on K by

$$\sum_p \frac{12p^{-\sigma_0}}{p-1},$$

which converges by comparison with $\sum_p p^{-1-\sigma_0}$. Using (4), the higher repetitions satisfy

$$\begin{aligned} \sum_p \sum_{n \geq 2} \frac{|c_{p,n}|}{n} p^{-n\sigma_0} &\leq 4 \sum_p \sum_{n \geq 2} \frac{(4p^{-\sigma_0})^n}{n} \\ &\ll \sum_p p^{-2\sigma_0} < \infty. \end{aligned}$$

Thus the logarithmic series converges normally. Its exponential is holomorphic and nonzero, proving Theorem 1.1.

References

- [1] Pierre de la Harpe. The fuglede–kadison determinant: Theme and variations. *Proceedings of the National Academy of Sciences*, 97(15):8263–8268, 2000.
- [2] Bent Fuglede and Richard V. Kadison. Determinant theory in finite factors. *Annals of Mathematics*, 55(3):520–530, 1952.